Probability, Statistics and Random Processes

Third Edition

T Veerarajan
Professor of Mathematics and Head
Science and Humanities Department
Sree Sowdambika College of Engineering
Aruppukottai, Tamil Nadu

Tata McGraw-Hill Publishing Company Limited
NEW DELHI

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Tata McGraw-Hill

Published by the Tata McGraw-Hill Publishing Company Limited,
7 West Patel Nagar, New Delhi 110 008.

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This edition can be exported from India only by the publishers,
Tata McGraw-Hill Publishing Company Limited.

ISBN 13: 978-0-07-066925-3
ISBN 10: 0-07-066925-2

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Published by the Tata McGraw-Hill Publishing Company Limited, 7 West Patel Nagar, New Delhi 110 008, typeset at Bukprint India, B-180A, Guru Nanak Pura, Laxmi Nagar-110 092 and printed at Adarsh Printers, C-50-51, Mohan Park, Naveen Shahdara, Delhi 110 032

Cover: SDR Printers

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PROBABILITY DISTRIBUTIONS

A **probability distribution** describes how the values of a random variable are distributed. It gives the probability that a random variable assumes a particular value or falls within a particular range.

Probability distributions are broadly classified into two categories:

1. **Discrete Probability Distributions**
2. **Continuous Probability Distributions**

Classification of Probability Distributions

Discrete Probability Distributions

A discrete probability distribution describes random variables that assume **countable values**.

Examples include:

- Binomial distribution
- Multinomial distribution
- Poisson distribution
- Hypergeometric distribution

Continuous Probability Distributions

Continuous probability distributions describe random variables that assume **continuous values**.

Examples include:

- Normal distribution
- Standard normal distribution
- Gamma distribution
- Exponential distribution
- Chi-square distribution
- Lognormal distribution
- Weibull distribution

Binomial Distribution

In many experiments, the outcome has only **two possible results**:

- Success
- Failure

An experiment consisting of repeated trials, each having two outcomes, is called a **Bernoulli process**. Each trial in such an experiment is called a **Bernoulli trial**.

Example

Consider firing bullets at a target.

Let the random variable

$$X = \text{number of successes in } n \text{ trials}$$

The random variable X follows a **binomial distribution** if the following conditions hold:

1. The experiment consists of **n identical trials**.
2. Each trial results in one of two mutually exclusive outcomes.
3. The probability of success in each trial is **p**, which remains constant.
4. All trials are **independent**.

Probability Mass Function of Binomial Distribution

The probability mass function of the binomial distribution is

$$P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, 2, \ldots, n$$

where

- p = probability of success
- $q = 1 - p$ = probability of failure
- n = number of trials
- x = number of successes

Constants of Binomial Distribution

Mean

$$\mu = np$$

Variance

$$\sigma^2 = npq$$

Standard Deviation

$$\sigma = \sqrt{npq}$$

Example

If the mean and variance of a binomial distribution are **4 and 4/3 respectively**, find $P(X \geq 1)$.

Solution

Mean

$$np = 4$$

Variance

$$npq = \frac{4}{3}$$

Substitute $np = 4$

$$4q = \frac{4}{3}$$

$$q = \frac{1}{3}$$

$$p = 1 - q = \frac{2}{3}$$

$$n \times \frac{2}{3} = 4$$

$$n = 6$$

Thus

$$X \sim Bin(6, \frac{2}{3})$$

Now

$$P(X \geq 1) = 1 - P(X = 0)$$

$$P(X = 0) = \binom{6}{0} \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^6$$

$$= \left(\frac{1}{3}\right)^6$$

$$= \frac{1}{729}$$

Therefore

$$P(X \geq 1) = 1 - \frac{1}{729}$$

$$= \frac{728}{729}$$

Poisson Distribution

The **Poisson distribution** is used to describe the number of occurrences of an event in a fixed interval of time or space.

Examples include:

- Number of accidents in a day
- Number of telephone calls received in a minute
- Number of defects in a manufactured product

Probability Function of Poisson Distribution

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

where

λ = mean number of occurrences.

Relationship between Binomial and Poisson Distribution

The Poisson distribution is a **limiting case of the binomial distribution** when

1. $n \to \infty$
2. $p \to 0$
3. $np = \lambda$ remains finite.

Example

A firm owns two cars and hires one every day. The number of demands for a car on each day follows a Poisson distribution with mean **1.5**.

Find the proportion of days

1. on which there is no demand
2. on which demand is refused.

Solution

For a Poisson distribution

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

Given

$$\lambda = 1.5$$

(i) No demand

$$P(X = 0) = e^{-1.5}\frac{(1.5)^0}{0!}$$

$$= e^{-1.5}$$

$$= 0.223$$

Number of days in a year

$$365 \times 0.223 \approx 81$$

Thus approximately **81 days** have no demand.

(ii) Demand refused

Demand is refused when

$$X > 2$$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$= 1 - [P(0) + P(1) + P(2)]$$

$$= 0.192$$

Number of days

$$365 \times 0.192 \approx 70$$

Thus demand is refused on about **70 days**.

Normal Distribution

The **normal distribution** is a continuous probability distribution widely used in statistics.

The probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

where

- μ = mean
- σ = standard deviation

Properties of Normal Distribution

1. Mean = Median = Mode
2. The curve is symmetrical about the mean
3. Total area under the curve is **1**

Standard Normal Distribution

If

$$X \sim N(\mu, \sigma^2)$$

then

$$Z = \frac{X - \mu}{\sigma}$$

follows the **standard normal distribution**

$$Z \sim N(0, 1)$$

Example

Mean = 8

Standard deviation = 4

Find

1. $P(5 \leq X \leq 10)$
2. $P(X \geq 5)$

Solution

$$Z = \frac{X - 8}{4}$$

For $X = 5$

$$Z = -0.75$$

For $X = 10$

$$Z = 0.5$$

$$P(5 \leq X \leq 10) = P(-0.75 \leq Z \leq 0.5)$$

Using standard normal tables

$$0.6915 - 0.2266$$

$$= 0.4649$$

Second part

$$P(X \geq 5) = P(Z \geq -0.75)$$

$$= 1 - 0.2266$$

$$= 0.7734$$

Normal Distribution Example

In a sample of **1000 cases**, the mean of a test is **14** and the standard deviation is **2.5**.

Find

1. Number of students scoring between **12 and 15**
2. Number scoring above **18**
3. Number scoring below **18**

Solution

$$Z = \frac{X - 14}{2.5}$$

For $X = 12$

$$Z = -0.8$$

For $X = 15$

$$Z = 0.4$$

$$P(12 \leq X \leq 15) = 0.6554 - 0.2119$$

$$= 0.4435$$

Number of students

$$1000 \times 0.4435 \approx 444$$

Students scoring above 18

$$Z = \frac{18 - 14}{2.5} = 1.6$$

$$P(X > 18) = 1 - 0.9452$$

$$= 0.0548$$

Number of students

$$1000 \times 0.0548 \approx 55$$

Students scoring below 18

$$P(X < 18) = 0.9452$$

Students

$$1000 \times 0.9452 \approx 945$$

Normal Distribution Example

The local corporation authorities in a certain city install **10,000 electric lamps** in the streets of the city with the assumption that the life of the lamps is normally distributed.

The lamps have:

Mean life

$$\mu = 1000 \text{ hours}$$

Standard deviation

$$\sigma = 200 \text{ hours}$$

Find

1. the number of lamps expected to fail in the **first 800 hours**
2. the number of lamps expected to fail **between 800 and 1200 hours**

Solution

Standard normal transformation

$$Z = \frac{X - \mu}{\sigma}$$

(i) Lamps failing before 800 hours

$$Z = \frac{800 - 1000}{200}$$

$$Z = -1$$

From the standard normal table

$$P(Z < -1) = 0.1587$$

Number of lamps

$$10000 \times 0.1587$$

$$= 1587$$

Thus **1587 lamps** are expected to fail within 800 hours.

(ii) Lamps failing between 800 and 1200 hours

For $X = 1200$

$$Z = \frac{1200 - 1000}{200}$$

$$Z = 1$$

$$P(-1 < Z < 1) = 0.6826$$

Number of lamps

$$10000 \times 0.6826$$

$$= 6826$$

Thus **6826 lamps** are expected to fail between 800 and 1200 hours.

Gamma Distribution

The **Gamma distribution** is a continuous probability distribution used in statistics and probability theory.

The **Gamma integral** is defined as

$$\Gamma(n) = \int_0^\infty x^{n-1} e^{-x} dx$$

For positive integers

$$\Gamma(n) = (n-1)!$$

Erlang Distribution

The Erlang distribution is a special case of the Gamma distribution where the parameter is an integer.

The probability density function is

$$f(x) = \frac{\lambda^k}{(k-1)!} x^{k-1} e^{-\lambda x}, \quad x > 0$$

where

- k = shape parameter
- λ = rate parameter

Example

In a certain city, the daily consumption of electric power (in millions of kWh) can be treated as a random variable having an **Erlang distribution** with parameters

$$\lambda = \frac{1}{2}$$

$$k = 3$$

If the power plant has a daily capacity of **12 million kWh**, find the probability that the power supply will be inadequate.

Solution

Let

$$X = \text{daily power consumption}$$

$$X \sim Erlang(k = 3, \lambda = \tfrac{1}{2})$$

The density function

$$f(x) = \frac{\lambda^k}{(k-1)!} x^{k-1} e^{-\lambda x}$$

Substitute

$$k = 3, \lambda = \frac{1}{2}$$

$$f(x) = \frac{(1/2)^3}{2!} x^2 e^{-x/2}$$

$$= \frac{1}{16} x^2 e^{-x/2}$$

Supply is inadequate when

$$X > 12$$

For Erlang distribution

$$P(X > x) = e^{-\lambda x} \left(1 + \lambda x + \frac{(\lambda x)^2}{2!}\right)$$

Substitute

$$\lambda = \frac{1}{2}, x = 12$$

$$\lambda x = 6$$

$$P(X > 12) = e^{-6}(1 + 6 + 18)$$

$$= 25e^{-6}$$

$$\approx 0.062$$

Thus the probability that supply is inadequate is **0.062**.

Example

If a company employs n salespersons, its gross sales (in rupees) follow an Erlang distribution with parameters

$$\lambda = \frac{1}{2}$$

$$k = 80n$$

If the sales cost is **Rs. 8000 per salesperson**, determine the number of salespersons required to **maximize expected profit**.

Solution

Let

$$X = \text{gross sales}$$

For Erlang distribution

$$E(X) = \frac{k}{\lambda}$$

Substitute

$$k = 80n, \lambda = \frac{1}{2}$$

$$E(X) = \frac{80n}{1/2}$$

$$= 160000n$$

Expected profit

$$y = \text{expected sales} - \text{sales cost}$$

$$y = 160000n - 8000n$$

Setting derivative conditions gives

$$n = 100$$

Thus **100 salespersons maximize the expected profit**.

Example

Consumer demand for milk in a locality per month follows a **Gamma distribution**.

Let

Average demand

$$a$$

Most likely demand

$$b$$

where

$$b < a$$

Find the **variance** of demand.

Solution

Gamma density function

$$f(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}$$

Mode

$$\frac{k-1}{\lambda} = b$$

Mean

$$\frac{k}{\lambda} = a$$

From mean

$$\lambda = \frac{k}{a}$$

Substitute into mode

$$\frac{k-1}{k/a} = b$$

Solving gives

$$k = \frac{a}{a-b}$$

Variance of Gamma distribution

$$Var(X) = \frac{k}{\lambda^2}$$

Substituting parameters

$$Var(X) = a(a-b)$$

Exponential Distribution

A random variable X is said to follow an **exponential distribution** if its probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

where

$\lambda > 0$ is called the **rate parameter**.

Properties of Exponential Distribution

Mean

$$E(X) = \frac{1}{\lambda}$$

Variance

$$Var(X) = \frac{1}{\lambda^2}$$

Moment generating function

$$M_X(t) = \frac{1}{1 - \frac{t}{\lambda}}$$

Applications of Exponential Distribution

The exponential distribution is useful in describing the **time taken for an event to occur**.

Examples include:

- Time between vehicle arrivals at a toll booth
- Time required to complete a questionnaire
- Distance between major defects in a highway
- Waiting time in queue systems

Memoryless Property

The exponential distribution has a special property called the **memoryless property**.

$$P(X > s + t | X > s) = P(X > t)$$

This means that the future probability does not depend on how much time has already passed.

Example

The lifetime of an electric bulb (in thousands of hours) follows an exponential distribution with rate

$$\lambda = \frac{1}{3}$$

The probability density function is

$$f(x) = \frac{1}{3}e^{-x/3}, \quad x > 0$$

The survival function is

$$P(X > t) = e^{-\lambda t}$$

(a) Probability that the bulb lasts more than 3 thousand hours

$$P(X > 3) = e^{-3/3}$$

$$= e^{-1}$$

$$= 0.3679$$

(b) Probability that the lifetime lies between 1.5 and 3 thousand hours

$$P(1.5 \leq X \leq 3) = P(X \leq 3) - P(X < 1.5)$$

$$= (1 - e^{-1}) - (1 - e^{-0.5})$$

$$= e^{-0.5} - e^{-1}$$

$$= 0.6065 - 0.3679$$

$$= 0.2386$$

(c) Conditional probability

$$P(X > 3.5 | X > 2.5)$$

Using memoryless property

$$P(X > 3.5 | X > 2.5) = P(X > 1)$$

$$= e^{-1/3}$$

$$= 0.7165$$

Example

The time required to repair a machine follows an exponential distribution with parameter

$$\lambda = \frac{1}{2}$$

Let X denote the repair time.

$$f(x) = \frac{1}{2}e^{-x/2}, \quad x > 0$$

(a) Probability that repair time exceeds 2 hours

$$P(X > 2) = e^{-2/2}$$

$$= e^{-1}$$

$$= 0.3679$$

(b) Conditional probability

Find

$$P(X \geq 10 | X > 9)$$

Using memoryless property

$$P(X \geq 10 | X > 9) = P(X > 1)$$

$$= e^{-1/2}$$

$$= 0.6065$$

Exercise Problems (Exponential Distribution)

Problem 1

The mileage obtained with a certain type of radial tyre follows an exponential distribution with mean **4000 km**.

Find the probability that a tyre lasts

1. at least 2000 km
2. at most 3000 km

Solution

Mean

$$\frac{1}{\lambda} = 4000$$

$$\lambda = \frac{1}{4000}$$

(i)

$$P(X \geq 2000) = e^{-2000/4000}$$

$$= e^{-0.5}$$

$$= 0.6065$$

(ii)

$$P(X \leq 3000) = 1 - e^{-3000/4000}$$

$$= 1 - e^{-0.75}$$

$$= 0.5270$$

Problem 2

The time for a camera to run without resetting follows an exponential distribution with mean **50 days**.

Find the probability that the camera

1. resets in less than 20 days
2. runs for at least 60 days

Solution

$$\lambda = \frac{1}{50}$$

(i)

$$P(X < 20) = 1 - e^{-20/50}$$

$$= 0.3297$$

(ii)

$$P(X \geq 60) = e^{-60/50}$$

$$= 0.3012$$

Problem 3

Length of telephone calls is exponentially distributed with mean **3 minutes**.

Find the probability that

1. waiting time exceeds 5 minutes
2. call lasts between 2 and 6 minutes

Solution

$$\lambda = \frac{1}{3}$$

(i)

$$P(X > 5) = e^{-5/3}$$

$$= 0.1889$$

(ii)

$$P(2 < X < 6) = e^{-2/3} - e^{-2}$$

$$= 0.3781$$

Problem 4

The lifetime of automobile batteries follows an exponential distribution with parameter

$$\lambda = 0.0001$$

Find

1. $P(X > 1200)$
2. $P(X > 1200 | X > 1000)$

Solution

(i)

$$P(X > 1200) = e^{-0.0001 \times 1200}$$

$$= 0.3012$$

(ii)

Using memoryless property

$$P(X > 1200 | X > 1000) = P(X > 200)$$

$$= e^{-0.02}$$

$$= 0.8187$$

Weibull Distribution

The **Weibull distribution** is widely used in reliability engineering and failure analysis.

The probability density function is

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^{\beta}}, \quad x > 0$$

where

- α = scale parameter
- β = shape parameter

Mean of Weibull Distribution

$$E(X) = \alpha^{-1/\beta}\Gamma\left(1 + \frac{1}{\beta}\right)$$

Example

The lifetime of a component follows Weibull distribution with parameters

$$\alpha = 0.2, \quad \beta = 0.5$$

Find the mean lifetime.

Solution

$$E(X) = \alpha^{-1/\beta}\Gamma\left(1 + \frac{1}{\beta}\right)$$

$$= 0.2^{-2}\Gamma(3)$$

$$= 25 \times 2$$

$$= 50$$

Thus the mean lifetime is **50 hours**.

Example

The service life of a semiconductor follows Weibull distribution with parameters

$$\alpha = 0.025, \quad \beta = 0.5$$

Find the probability that it functions for more than **4000 hours**.

Solution

Survival function

$$P(X > t) = e^{-\alpha t^{\beta}}$$

$$P(X > 4000) = e^{-0.025(4000)^{0.5}}$$

$$= e^{-1.581}$$

$$= 0.2057$$

Binomial Exercise

Suppose there are **12 multiple choice questions** in a quiz.

Each question has **5 possible answers** and only **one correct answer**.

Find the probability of getting **four or fewer correct answers** when answers are chosen randomly.

Solution

$$p = \frac{1}{5}$$

$$n = 12$$

$$P(X \leq 4) = 0.9274$$

Binomial Distribution Example

10% of screws produced by a machine are defective.

From **20 screws selected randomly**, find the probability that

1. exactly 2 are defective
2. at least 2 are defective
3. between 1 and 3 are defective

Solution

$$n = 20$$

$$p = 0.1$$

(i)

$$P(X = 2) = 0.2852$$

(ii)

$$P(X \geq 2) = 0.6083$$

(iii)

$$P(1 \leq X \leq 3) = 0.7344$$